

BOLTENHAGEN, GermanyWMO: **101610**Lat: **54.0028N**Lon: **11.1908E**Elev: **16**StdP: **101.13**Time Zone: **1.00 (EUC)**Period: **82-03**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.2	-7.3	-13.2	1.2	-8.7	-11.0	1.5	-7.1	16.5	3.8	15.1	3.3	3.9	220	0.689

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	25.1	19.2	23.2	18.2	21.7	17.5	19.8	23.7	18.9	22.4	18.0	21.1	4.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	13.1	22.1	17.4	12.5	21.0	16.6	11.8	20.1	56.7	23.8	53.5	22.5	50.8	21.2	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.2	12.5	11.2	-10.5	30.1	5.4	2.9	-14.4	32.2	-17.5	34.0	-20.6	35.6	-24.5	37.8		
(5)			-10.9	22.2	5.3	1.0	-14.7	23.0	-17.8	23.6	-20.8	24.2	-24.6	24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	1.3	1.0	4.0	6.9	11.4	14.7	17.2	17.0	13.8	10.1	5.3	2.9		
	DBStd	6.62	4.91	4.26	3.52	3.05	2.95	2.85	2.56	2.59	2.31	3.13	3.08	3.44		
	HDD10.0	1230	269	253	186	101	18	1	0	0	1	37	143	221		
	HDD18.3	3505	528	486	443	342	216	114	54	58	138	256	392	479		
	CDD10.0	809	0	1	1	9	62	142	222	217	115	39	2	1		
	CDD18.3	42	0	0	0	0	1	5	18	17	2	0	0	0		
	CDH23.3	167	0	0	0	0	2	16	67	74	8	0	0	0		
	CDH26.7	32	0	0	0	0	0	2	12	17	1	0	0	0		
(14) Wind	WSAvg	5.5	6.8	6.3	5.7	5.1	4.9	4.6	4.8	4.6	5.2	5.6	6.0	6.3		
(15) Precipitation	PrecAvg	632	51	41	37	32	52	64	64	75	57	50	52	56		
	PrecMax	829	105	85	106	69	108	149	147	192	158	118	125	130		
	PrecMin	423	0	3	6	1	10	4	8	19	20	18	6	19		
	PrecStd	107	26	20	21	17	25	32	36	43	33	20	30	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.3	13.7	15.3	19.9	23.1	25.4	28.2	28.5	24.1	20.7	13.4	11.9		
		MCWB	9.7	11.2	11.0	12.8	16.8	19.3	20.2	19.5	18.9	17.3	11.4	10.0		
	2%	DB	9.2	10.6	12.1	15.9	19.5	22.5	25.0	25.2	20.9	17.5	11.8	10.6		
		MCWB	7.8	8.6	9.2	11.0	15.2	18.0	19.5	19.1	16.4	14.5	10.2	9.2		
	5%	DB	8.3	7.9	10.0	13.5	17.3	20.9	23.1	22.8	18.9	15.6	10.6	9.2		
		MCWB	7.0	6.5	7.8	10.0	14.0	17.1	18.5	18.1	15.5	13.2	9.4	8.1		
	10%	DB	7.1	6.3	8.6	11.5	15.9	19.5	21.5	21.2	17.5	14.3	9.4	7.8		
		MCWB	5.9	5.0	6.8	8.9	13.0	16.0	17.6	17.4	14.5	12.2	8.3	6.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	11.4	11.5	13.8	17.2	20.0	21.4	21.1	18.9	17.1	12.1	10.4		
		MCDB	10.8	13.5	14.7	17.6	22.0	23.8	26.4	25.9	23.5	20.3	13.1	11.6		
	2%	WB	8.1	8.6	9.6	11.6	15.7	18.7	19.9	19.8	17.3	15.0	10.5	9.3		
		MCDB	8.9	10.2	11.6	15.3	18.9	21.8	24.0	23.6	19.8	16.9	11.5	10.4		
	5%	WB	7.1	6.6	8.2	10.3	14.5	17.6	18.9	18.7	15.9	13.5	9.5	8.1		
		MCDB	8.2	7.8	9.9	13.0	17.0	20.3	22.4	22.2	18.2	15.1	10.5	9.1		
	10%	WB	6.0	5.1	6.9	9.1	13.3	16.4	18.0	17.8	14.8	12.5	8.5	6.8		
		MCDB	7.1	6.2	8.4	11.2	15.6	19.0	21.0	20.8	17.0	14.0	9.5	7.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	4.3	5.0	6.0	6.1	6.5	7.1	7.2	6.2	5.4	4.2	3.7		
		MCDBR	4.6	6.2	7.8	9.9	8.8	8.4	9.6	9.6	8.5	7.1	5.6	4.8		
	5% WB	MCWBR	4.2	5.1	5.7	6.2	5.6	5.5	5.7	5.4	5.5	5.1	5.0	4.5		
		MCWBR	4.6	6.4	7.6	9.1	8.1	8.1	8.8	8.4	7.7	6.5	5.3	4.6		
(40) Clear-Sky Solar Irradiance	taub		0.319	0.340	0.373	0.410	0.402	0.399	0.419	0.415	0.385	0.360	0.338	0.311		
		taud	2.311	2.315	2.255	2.206	2.291	2.323	2.288	2.311	2.378	2.410	2.367	2.307		
	Ebn at Noon		630	744	796	814	845	850	823	802	781	715	604	559		
		Edh at Noon	59	80	105	126	122	120	122	113	93	73	56	49		
(44) All-Sky Solar Radiation	RadAvg		0.53	1.18	2.42	4.07	5.16	5.45	5.11	4.17	3.04	1.63	0.68	0.39		
	RadStd		0.07	0.18	0.23	0.53	0.61	0.39	0.66	0.31	0.32	0.21	0.08	0.03		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-145	N/A	N/A

Nomenclature: See separate page